AIM

To examine the various types of cables used in computer networks and study their construction, performance, benefits, limitations, and common uses.

OBJECTIVE

The objectives of this study are:

- To identify the major types of network cables.
- To understand how different cables transmit data.
- To compare their speed, distance, cost, and interference resistance.
- To study the applications of copper and fibre-based cables.
- To determine which cable is suitable for different networking environments.

INTRODUCTION

A network cable is a physical medium that carries information between computers, switches, routers, servers, and other communication devices. The choice of cable depends on requirements such as network speed, transmission distance, installation cost, bandwidth, and the amount of electrical interference present in the surroundings.

Network cables can broadly be classified into **copper cables** and **fibre optic cables**. Copper cables transmit electrical signals, whereas fibre optic cables transmit information using pulses of light.

The commonly used network cables are:

1. Unshielded Twisted Pair (UTP)
2. Shielded Twisted Pair (STP)
3. Coaxial Cable
4. Fibre Optic Cable

1. UNSHIELDED TWISTED PAIR (UTP) CABLE

UTP cable is made up of multiple pairs of insulated copper wires. The wires in each pair are twisted around one another. This twisting helps reduce interference between the wire pairs and makes the cable suitable for Ethernet networks.

UTP is widely used because it is inexpensive, easy to handle, and simple to terminate using standard network connectors.

Common UTP Categories

Category	Approximate Maximum Speed	Typical Use
Cat 3	10 Mbps	Older Ethernet and telephone systems
Cat 5	100 Mbps	Fast Ethernet
Cat 5e	1 Gbps	Gigabit Ethernet
Cat 6	Up to 10 Gbps*	High-speed LAN connections
Cat 6a	Up to 10 Gbps	10 Gigabit Ethernet

*Cat 6 can support 10 Gbps over shorter specified distances.

Benefits

- Relatively inexpensive.
- Simple to install and terminate.
- Flexible and lightweight.
- Easily available.
- Suitable for most office and home networks.

Limitations

- More vulnerable to external electromagnetic interference than shielded cables.
- Copper Ethernet links have distance limitations.
- Performance can be affected by poor installation or excessive bending.
- Provides less noise protection in electrically noisy environments.

Uses

- Home and office LANs.
- Ethernet networks.
- Connections between computers and switches.
- Network outlets and patch panels.
- General-purpose data communication.

2. SHIELDED TWISTED PAIR (STP) CABLE

Shielded Twisted Pair cable is similar to UTP but includes a conductive metallic shield around the individual pairs, the cable as a whole, or both. This additional shielding reduces the effect of electromagnetic and radio-frequency interference.

STP is especially useful where networking equipment is installed close to motors, generators, electrical machinery, or other sources of interference.

Common Categories

Category	Approximate Maximum Speed	Typical Use
Cat 6	Up to 10 Gbps*	High-speed Ethernet
Cat 6a	Up to 10 Gbps	10G Ethernet
Cat 7	Up to 10 Gbps	High-performance structured cabling

*The supported speed depends on cable length, installation conditions, and the Ethernet standard.

Benefits

- Better protection against electromagnetic interference.
- Provides more stable communication in noisy environments.
- Suitable for high-performance networks.
- Helps reduce unwanted signal interference.
- Available in high-speed cable categories.

Limitations

- Costs more than comparable UTP cables.
- More difficult to install and terminate.
- Heavier and less flexible.
- Shielding must be correctly bonded and grounded where required.

Uses

- Industrial networking environments.
- Data centres and server rooms.
- Networks located near electrical equipment.
- High-speed Ethernet installations.
- Areas where electromagnetic interference is a concern.

3. COAXIAL CABLE

Coaxial cable contains a central conductor surrounded by an insulating dielectric layer, a conductive shield, and an external protective jacket. The shield helps protect the signal from external interference.

Although twisted-pair and fibre cables are more common in modern computer LANs, coaxial cable continues to be important in television, broadband, CCTV, and other communication systems.

Common Coaxial Cable Types

Cable Type	Common Application
RG-6	Cable television and broadband connections
RG-59	CCTV and video applications
RG-11	Longer cable-TV and broadband runs

Data Transmission

The data-carrying capability of coaxial cable varies considerably according to the cable design and communication technology. Older Ethernet implementations such as 10BASE2 and 10BASE5 operated at **10 Mbps**, while modern broadband systems using coaxial infrastructure can provide much higher data rates.

Benefits

- Good resistance to external interference.
- Strong and durable construction.
- Suitable for relatively long cable runs.
- Provides reliable signal transmission.
- Useful for both communication and video signals.

Limitations

- Thicker than most twisted-pair cables.
- Less convenient to route through confined spaces.
- Installation can be more difficult.
- Not normally the first choice for modern structured Ethernet cabling.

Uses

- Cable television systems.
- Broadband internet infrastructure.
- CCTV and video systems.
- Radio-frequency communication.
- Certain legacy computer networks.

4. FIBRE OPTIC CABLE

Fibre optic cable uses extremely thin strands of glass or plastic to transmit information as pulses of light. Unlike copper cables, fibre does not carry data as an electrical signal.

Fibre optic technology provides very high bandwidth, excellent resistance to electromagnetic interference, and the ability to transmit data over long distances.

Types of Fibre Optic Cable

Type	Main Characteristics	Typical Applications
Single-Mode Fibre (SMF)	Carries light through a very small core and is designed for long-distance transmission	Telecommunications, WANs, backbone networks
Multi-Mode Fibre (MMF)	Has a larger core and supports multiple light paths	Data centres, LANs, campus networks

Data Transmission

Fibre optic networks can operate at extremely high speeds. Depending on the fibre type, transceivers, network equipment, and transmission standard, modern systems can support **100 Gbps, 400 Gbps, and higher data rates**.

Benefits

- Extremely high bandwidth.
- Supports very long transmission distances.
- Not affected by electromagnetic interference.
- Low signal attenuation over suitable distances.
- Provides good resistance to electrical interference.
- Suitable for high-speed backbone and data-centre connections.
- Difficult to intercept compared with conventional copper cabling.

Limitations

- Installation can be more expensive.
- Requires specialised tools and equipment.
- Splicing and termination require trained personnel.
- Fibre strands need careful handling to avoid damage.
- Optical transceivers and related equipment can increase system cost.

Uses

- Internet backbone infrastructure.
- Telecommunication networks.
- Data centres.
- Campus and metropolitan networks.
- Long-distance WAN connections.
- High-speed network links.

COMPARISON OF NETWORK CABLES

Feature	UTP	STP	Coaxial	Fibre Optic
Transmission Medium	Copper	Shielded Copper	Copper	Glass/Plastic Fibre
Signal Type	Electrical	Electrical	Electrical/RF	Light
Typical Modern Ethernet Speed	Up to 10 Gbps	Up to 10 Gbps	Mostly legacy Ethernet; broadband varies	100 Gbps and above possible
EMI Resistance	Low to Moderate	High	High	Excellent
Flexibility	High	Moderate	Low	Moderate
Installation	Easy	Moderate	Moderate	More specialised
Relative Cost	Low	Medium	Medium	High
Transmission Distance	Limited	Limited	Moderate	Very long
Common Use	Home/office LAN	Industrial/high-interference LAN	TV, broadband, CCTV	Backbone, WAN, data centres

OBSERVATION

From the study, it can be observed that no single cable is ideal for every networking requirement. **UTP** is generally preferred for ordinary LAN installations because it is economical and easy to install. **STP** is more appropriate in environments where electrical interference is significant. **Coaxial cable** remains useful for television, broadband, CCTV, and specialised communication systems. **Fibre optic cable** is the preferred choice for very high-speed and long-distance communication.

CONCLUSION

Different network cables have different characteristics and applications.
Copper-based cables such as UTP, STP, and coaxial cables are useful for short- and medium-distance communication, while fibre optic cables are better suited for high-bandwidth and long-distance networks.

The selection of a network cable should therefore be based on **required speed, transmission distance, interference level, installation conditions, reliability, and overall cost**. Understanding these differences helps in designing an efficient and suitable computer network.